

Aryan Kakkar

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EDUCATION

Wilfrid Laurier University

Sep. 2023 – Aug. 2027

Honours Bachelor of Science in Computer Science (Co-op), Minor in Economics

Waterloo, ON

- Concentration in Big Data Systems
- Relevant Coursework: Probability & Statistics (R), Linear Algebra, Algorithms, Data Structures, Machine Learning, Databases, Data Visualization, Canadian Banking & Financial Systems

EXPERIENCE

Research Assistant

Jan. 2026 – Present

Wilfrid Laurier University

Waterloo, ON

- Built and evaluated **Random Forest, Logistic Regression, LSTM, and BiGRU** models for V2G discharge prediction, finding that **engineered features captured most predictive signal**
- Developed an end-to-end **ML pipeline in Python, PyTorch, and Pandas** for ingesting, aligning, and modeling **72K+ EV charging sessions and 8.7K hourly grid observations**
- Engineered a rule-based labeling framework combining **EV availability aggregates** and **CAISO grid demand thresholds** to generate **V2G/G2V supervised labels** across **8,735 hourly observations**, enabling all downstream model training
- Deployed a full **EVERest** software stack via **Docker** and studied **OCPP** and **ISO 15118** protocol standards to validate real-world V2G bidirectional charging assumptions and ground the modeling approach in hardware behavior

Machine Learning Engineer Intern

Sep. 2025 – Dec. 2025

Multiverse Computing

Toronto, ON

- Built and maintained **end-to-end production data pipelines** to ingest, preprocess, and aggregate multi-source football performance datasets spanning **700+ academy and first-team players using Python**
- Implemented **Monte Carlo simulation pipelines** with Gaussian uncertainty modeling to predict **player performance, health, wellness scores, and injury risk across 700+ longitudinal player records**
- Led evaluation of **XGBoost, Random Forest, Linear Regression, and Quantile Regression** models for **football youth team performance prediction**, benchmarking across shared pipelines from data ingestion through model evaluation
- Engineered an **LLM-based multimodal edge deployment**, integrating **VAD, STT, and TTS** components to enable real-time voice-interactive workflows on-device
- Developed **Streamlit dashboards** to visualize and compare model outputs across interns, performed **data cleaning and null imputation**, and compiled a **project report** documenting methodology and results for supervisor review

Director of Logistics

Sep. 2025 - Dec. 2025

Laurier Cricket Club

Waterloo, ON

- Directed end-to-end planning and execution of **10+ events and matchdays** as Director of Logistics, coordinating across departments to streamline operations and improve cross-functional workflows
- Identified process inefficiencies and implemented **Notion and Slack** to centralize communications and automate day-to-day club tasks, supporting **VP Operations** in delivering organized, scalable club management

Project Developer

Sep. 2024 - Sep. 2025

Laurier Cybersecurity Club

Waterloo, ON

- Built **analytics features** for a **full-stack platform**, enabling **real-time tracking of event and user metrics** through integrated backend APIs and front-end dashboards
- Designed and implemented **data flows** between backend services and visualization layers, ensuring accurate reporting of **operational metrics**

PROJECTS

NetTrace - Graph-Based Fraud Detection | Hackathon

Mar. 2026

- Modeled 79,650 financial accounts as a directed transaction graph and trained a **GraphSAGE** node classifier achieving **AUC 0.9237** and **81.1% precision**, improving from a 0.866 AUC baseline
- Built a **3-signal ensemble** combining supervised GNN, **Isolation Forest** anomaly detection, and **Louvain** community scoring to surface \$9.8B in fraud-associated volume across 10,074 confirmed accounts
- Deployed an interactive **Streamlit** dashboard with tunable risk threshold, flagged account explorer, and financial impact visualizations over a full 4-layer pipeline on the **PaySim** dataset (6.3M transactions)

Personal Portfolio

Oct. 2025 - Dec 2025

- Designed and built a **responsive web application** using **React, JavaScript, and HTML/CSS** to showcase projects, experience, and technical skills with a focus on clarity, usability, and accessibility
- Improved performance and user experience through iterative testing, structured component design, and **version-controlled development** using **Git** and **GitHub**, incorporating accessibility-focused UI improvements

Budget Planner

Apr. 2025 - Jul. 2025

- Built an **interactive financial dashboard** using **React, JavaScript, and CSS** to analyze spending patterns and simulate savings scenarios based on user-defined inputs
- Implemented **data aggregation logic** in **JavaScript** to compute monthly totals and category level breakdowns, and created clear visual summaries using **Chart.js**, with version control and collaboration managed by **Git** and **GitHub**

TECHNICAL SKILLS

Languages: Python, SQL, R, Java, C, JavaScript, HTML, CSS, TypeScript, VBA, PowerShell

Data & Modeling: NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow, PyTorch Geometric, NetworkX

Visualization & APIs: Matplotlib, Streamlit, Chart.js, FastAPI, Flask

Developer Tools: Git/GitHub Actions, JupyterLab, VSCode, JetBrains, Jest, Figma